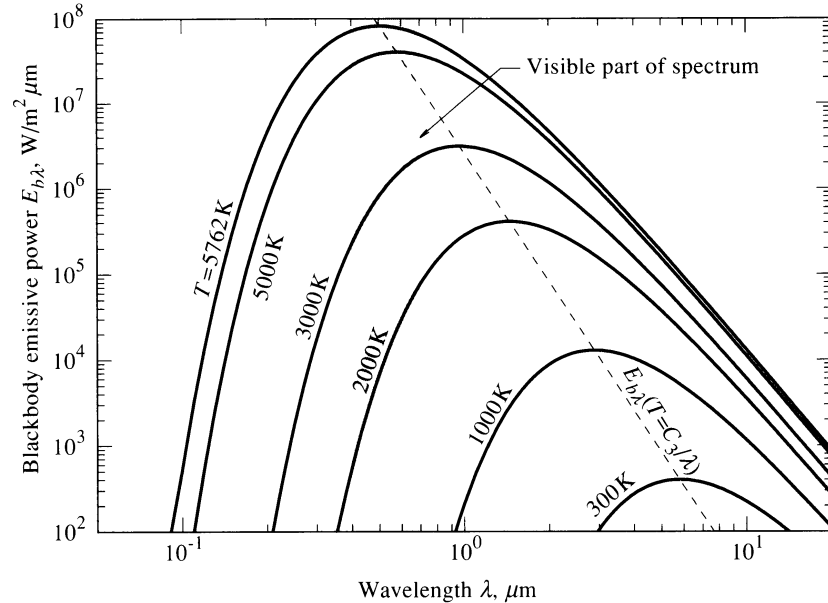


Blackbody Radiation Spectrum



Plank distribution, $C_3 = 2897.8 [\mu\text{m K}]$

$$E_b = n^2 \sigma T^4 \quad \text{Stefan-Boltzmann Law}$$

$$\sigma = \frac{2C_1\pi^5}{15C_2^4} = 5.6704 \times 10^{-8} \left[\frac{\text{W}}{\text{m}^2 \text{K}^4} \right]$$

$$E_{b\nu} d\nu = E_{b\lambda} d\lambda \left[\frac{\text{W}}{\text{m}^2} \right]$$

$$E_{b\nu}(T, \nu) = \frac{2\pi h\nu^3 n^2}{C_0 [\exp(\frac{h\nu}{kT}) - 1]} \left[\frac{\text{W}}{\text{m}^2 \text{Hz}} \right]$$

$$E_{b\lambda}(T, \lambda) = \frac{2\pi h C_0^2}{n^2 \lambda^5 [\exp(\frac{hC_0}{n k T}) - 1]} \left[\frac{\text{W}}{\text{m}^2 \mu\text{m}} \right]$$

$$\frac{E_{b\lambda}}{n^3 T^5} = \frac{2\pi C_1}{(n\lambda T)^5 (\exp(\frac{C_2}{n\lambda T}) - 1)}$$

$$C_1 = hC_0^2 = 0.59552 \times 10^8 \left[\frac{\text{W} \mu\text{m}^4}{\text{m}^2} \right]$$

$$C_2 = \frac{hC_o}{k} = 14388 (\mu\text{m K})$$

$n\lambda T$ [$\mu\text{m K}$]	η/nT [cm^{-1}/K]	$E_{b\lambda}/n^3T^5$ [$\text{W}/\text{m}^2\mu\text{m K}^5$]	$E_{b\eta}/nT^3$ [$\text{W}/\text{m}^2\text{cm}^{-1}\text{K}^3$]	$f(n\lambda T)$
1000	10.0000	0.02110×10^{-11}	0.00211×10^{-8}	0.00032
1100	9.0909	0.04846	0.00586	0.00091
1200	8.3333	0.09329	0.01343	0.00213
1300	7.6923	0.15724	0.02657	0.00432
1400	7.1429	0.23932	0.04691	0.00779
1500	6.6667	0.33631	0.07567	0.01285
1600	6.2500	0.44359	0.11356	0.01972
1700	5.8824	0.55603	0.16069	0.02853
1800	5.5556	0.66872	0.21666	0.03934
1900	5.2632	0.77736	0.28063	0.05210
2000	5.0000	0.87858	0.35143	0.06672
2100	4.7619	0.96994	0.42774	0.08305
2200	4.5455	1.04990	0.50815	0.10088
2300	4.3478	1.11768	0.59125	0.12002
2400	4.1667	1.17314	0.67573	0.14025
2500	4.0000	1.21659	0.76037	0.16135
2600	3.8462	1.24868	0.84411	0.18311
2700	3.7037	1.27029	0.92604	0.20535
2800	3.5714	1.28242	1.00542	0.22788
2900	3.4483	1.28612	1.08162	0.25055

$$f(n\lambda T) =$$

Fraction of radiation energy in
the range $\{0 : \lambda T\}$

$$\text{i.e., } \int_0^\lambda E_{b\lambda} d\lambda / \int_0^\infty E_{b\lambda} d\lambda$$

5 terms of the following series
give good results

$$f_{0-\lambda T} = \frac{15}{\pi^4} \sum_{n=1}^{\infty} \left[\frac{e^{-n\xi}}{n} \left(\xi^3 + \frac{3\xi^2}{n} + \frac{6\xi}{n^2} + \frac{6}{n^3} \right) \right] \quad \text{where } \xi = \frac{C_2}{\lambda T}$$